



Tribhuvan University

Faculty of Humanities and Social Sciences

COURIER MANAGEMENT SYSTEM

A PROJECT REPORT

Submitted to

Department of Computer Application

Everest Innovative College

Soalteemode, Kalimati

In partial fulfillment of the requirements for the Bachelors in Computer Application

Submitted by

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Bachelor in Computer Applications (BCA)

SUPERVISOR'S RECOMMENDATION

I hereby recommend that this project prepared under my supervision by **Raja Ram Bhurtel** entitled “**Courier Management System**” in the Partial Fulfillment of requirement for the degree of Bachelor in Computer Application is recommended for that final evaluation.

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LETTER OF APPROVAL

This is to certify that this project prepared by **Raja Ram Bhurtel** entitled “**Courier Management System**” in the Partial Fulfillment of requirement for the degree of Bachelor in Computer Application has been evaluated. In our opinion it is satisfactory in the scope and quality as a project for the required degree.

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Yours sincerely,

Raja Ram Bhurtel

ABSTRACT

In the ever-evolving realm of logistics and transportation, the need for a transformative solution to address the challenges of manual handling, tracking deficiencies, and transparency gaps during product transit has become imperative. This report explores the fundamental issue of the absence of a comprehensive system for automating shipping operations and providing real-time tracking to consumers. Introducing the Courier Management System (CMS), an online application designed to empower cargo management professionals by streamlining item reception, secure delivery, and real-time journey monitoring. The report emphasizes CMS's pivotal role in modernizing and enhancing courier management, fostering customer relationships, enabling efficient tracking, and facilitating seamless stakeholder communication. Through case studies and analysis, it highlights CMS's potential to optimize operations and elevate customer satisfaction, offering valuable insights for industry professionals and stakeholders eager to embrace efficient and transparent courier management. This project utilizes PHP, JavaScript, and MYSQL, incorporating algorithms such as the Vehicle Routing Problem and distance-based price calculation using Haversine Formula. The system operates as an online application, accessible via the internet or a company's local area network, thereby requiring an internet connection for user access.

Keywords: Courier Management, Real-time Tracking, Customer Satisfaction, Vehicle Routing Problem, Haversine Formula

TABLE OF CONTENTS

SUPERVISOR’S RECOMMENDATION	i
LETTER OF APPROVAL.....	ii
ACKNOWLEDGEMENT.....	iii
ABSTRACT.....	iv
TABLE OF CONTENTS	v
LIST OF ABBREVIATIONS	vii
LIST OF FIGURES	viii
LIST OF TABLES	ix
CHAPTER 1 INTRODUCTION	1
1.1 Introduction	1
1.2 Problem Statement	2
1.3 Objectives	2
1.4 Scope and Limitation.....	2
1.4.1 Scope	2
1.4.2 Limitations.....	3
1.5 Development Methodology	3
1.6 Report Organization	5
CHAPTER 2 BACKGROUND STUDY AND LITERATURE REVIEW	6
2.1 Background Study	6
2.2 Literature Review	7
CHAPTER 3 SYSTEM ANALYSIS AND DESIGN	9
3.1 System Analysis	9
3.1.1 Requirement Analysis.....	9
3.1.2 Feasibility Analysis	12
3.1.3 Object Modelling using Class and Object Diagrams.....	14
3.1.4 Dynamic Modelling using Sequence Diagrams	16
3.1.5 Process Modelling using Activity Diagrams	19

3.2 System Design	21
3.2.1. Refinement of Class diagram	21
3.2.2. Component Diagrams	22
3.2.3. Component Diagrams	23
3.3 Algorithm Details	23
3.3.1. Vehicle Routing Problem Algorithm.....	23
3.3.2. Algorithm for calculating price based on distance	24
3.3.3. Algorithm for scanning the QR code from the uploaded image.....	25
CHAPTER 4 IMPLEMENTATION AND TESTING	26
4.1 Implementation.....	26
4.1.1. Tools Used (CASE tools, Programming language, Database platforms).....	26
4.1.2. Implementation Details of Modules (Description of procedures / Functions / Methods).....	27
4.2 Testing.....	27
4.2.1. Test Cases for Unit Testing	27
4.2.2. Test Cases for System Testing.....	31
4.3 Result Analysis.....	32
CHAPTER 5 CONCLUSION AND FUTURE RECOMMENDATIONS.....	37
5.1 Lesson Learnt / Outcome	37
5.2 . Conclusion.....	38
5.3 Future Recommendations.....	39
REFERENCES.....	40
APPENDICES: SYSTEM SCREENSHOTS	

LIST OF ABBREVIATIONS

CMS	Courier Management System
VRP	Vehicle Routing Problem
DHL	Dalsey, Hillblom and Lynn
POD	Proof of Delivery
ROI	Region of Interest
PDF	Portable Document Format
KTM	Kathmandu
PIECES	Performance Information Economics Control Efficiency Service
XAMPP	Cross-platform, Apache, Maria DB(MySQL), PHP and Perl

LIST OF FIGURES

Figure 1.1: Waterfall Model	4
Figure 3.1: Use Case Diagram of Courier Management System.....	12
Figure 3.2: Gantt chart for Courier Management System.....	14
Figure 3.3: Class Diagram for Courier Management System.....	15
Figure 3.4: Object Diagram for Courier management System	15
Figure 3.5: Squence Diagram for login to CMS	16
Figure 3.6: Squence Diagram for booking request in CMS.....	16
Figure 3.7: Squence Diagram for booking management in CMS.....	17
Figure 3.8: Squence Diagram for user management in CMS	18
Figure 3.9: Squence Diagram for booking status in CMS	18
Figure 3.10: Squence Diagram for report management in CMS	19
Figure 3.11: User Activity Diagram for Courier Management System.....	19
Figure 3.12: Admin Activity Diagram for Courier Management System	20
Figure 3.13: Delivery person activity diagram for Courier Management System.....	20
Figure 3.14: Refinement of class diagram for Courier Management System.....	21
Figure 3.15: Component Diagram of Admin for Courier Management System	22
Figure 3.16: Component Diagram of Admin for Courier Management System	22
Figure 3.17: Deployment Diagram for Courier Management System.....	23

LIST OF TABLES

Table 3.1: Gantt chart Table for Courier Management System.....	13
Table 4.1: Test case for User Login of Courier Management System.....	27
Table 4.2: Test case for Booking of Courier Management System.....	28
Table 4.3: Test case for Create Manifest of Courier Management System	28
Table 4.4: Test case for Create Delivery of Courier Management System	28
Table 4.5: Test case for Create User of Courier Management System.....	29
Table 4.6: Test case for Create Merchandise of Courier Management System.....	29
Table 4.7: Test case for Create Consignee of Courier Management System	29
Table 4.8: Test case for Create Shipper of Courier Management System.....	30
Table 4.9: Test case for Check Status of Courier Management System.....	30
Table 4.10: Test case for Report Generation of Courier Management System	31
Table 4.11: System test case for Courier Management System.....	31
Table 4.12: Dataset parameters for distance-based price calculation	32
Table 4.13: Obtained Result	33
Table 4.14: Dataset for Vehicle Routing Problem.....	34

CHAPTER 1

INTRODUCTION

1.1 Introduction

The Courier Management System is currently being developed as an automated solution that is designed to streamline and enhance the operations of courier companies. It is actively addressing the challenges faced by traditional courier systems by providing a comprehensive platform for efficient order processing, real-time package tracking, dynamic delivery route optimization, and seamless customer communication. By leveraging advanced technologies and cutting-edge algorithms, the system is automating various processes, eliminating manual tasks, and significantly reducing the likelihood of errors. Integration of these key functionalities into a centralized platform empowers courier companies to improve their operational efficiency, enhance customer satisfaction, and gain a competitive edge in the industry.

As the Courier Management System continues to evolve, it is bringing numerous benefits to courier companies. Improved operational efficiency is being achieved through streamlined processes, reduced manual errors, and optimized resource allocation. The system's advanced package tracking capabilities ensure accurate and real-time updates for customers, leading to enhanced satisfaction and trust. Furthermore, the dynamic delivery route optimization algorithm is continuously analyzing real-time data such as traffic conditions and package volumes, allowing courier companies to maximize efficiency, minimize travel time, and reduce costs.

The successful implementation of the Courier Management System will transform the entire operational landscape of courier companies. It will enable them to meet the ever-increasing demands of customers in a fast-paced and competitive market. By improving service quality, increasing efficiency, and providing better visibility into the delivery process, courier companies will gain a significant competitive advantage. The Courier Management System holds the potential to reshape the courier industry, driving sustainable growth, and positioning companies as leaders in the field.

1.2 Problem Statement

The current courier systems suffer from several limitations such as manual order processing, lack of real-time package tracking, inefficient delivery routes, and time-consuming administrative tasks. These limitations result in delays, inaccuracies, and poor customer experiences. Therefore, there is a need for an automated Courier Management System to overcome these challenges and improve the effectiveness of courier operations.

The major problems include:

- Manual Order Processing
- Inefficient Package Tracking
- Ineffective Delivery Route Planning

These challenges highlight the need for an automated Courier Management System that addresses these issues and provides a streamlined, efficient, and transparent solution. By automating order processing, implementing robust package tracking, optimizing delivery routes and improving customer communication, the Courier Management System aims to overcome these challenges and enhance the overall efficiency and effectiveness of courier operations.

1.3 Objectives

The objective is to carry out this project with the quality requirements as mentioned below:

- To increase order processing and reducing delivery time by route optimization.
- To integrate a customer self-service portal, allowing customers to track their shipments independently.

1.4 Scope and Limitation

1.4.1 Scope

The system is the integration of appropriate technologies to help managers manage information. Technologies that can take advantage of abundant domestic resources and save capital and labor are considered appropriate. This project deals with maintaining booking details, courier arrival details, courier non-delivery details, courier return details etc. The main purpose of this project is to computerize the maintenance of courier management.

1.4.2 Limitations

There are some criteria that may not be fulfilled by our application implemented. Some of such limitations of our project are mentioned below:

- Unemployment: some of the workers expressed fear that they will be laid off due to the introduction of computerized systems.
- Training cost: the employees will be trained on the effective use of the courier system which will cost the company both time and finances.
- Maintenance cost: the system will require modification due to new user requirements, upgrades and installations. However, the courier management system for the transport and cargo sector is worth undertaking as the benefits it will introduce are immense.

1.5 Development Methodology

This system is designed with the series of processes starting with requirement analysis, design, implementation, testing and maintenance. During requirement analysis, all the functional and non-functional requirement are analyzed and system is developed according to the requirement then designing of the system is carried out. After the design process, coding and development part is started then after integrating the system there is testing of the system. If the testing is positive, then system is implemented otherwise some maintenance is done and system come in operation.

The Waterfall Model was the first Process Model to be introduced which referred to as a linear-sequential life cycle model. [1] The whole process of software development is divided into separate phases in waterfall approach. In Waterfall model the outcome of one phase acts as the input for the next phase sequentially. [2] So there is no overlapping between the phases which makes it easy and simple to follow.

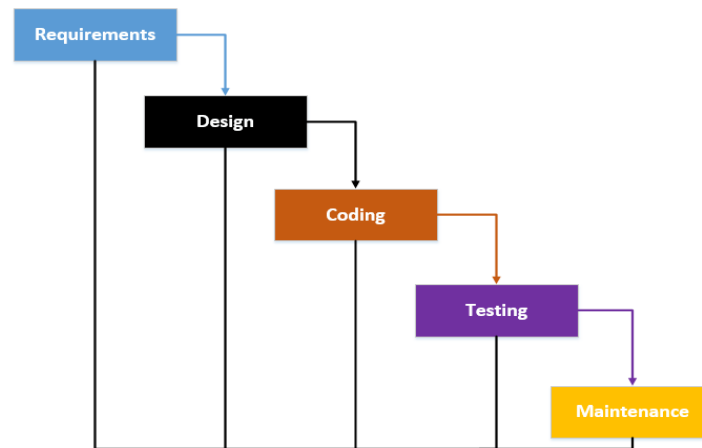


Figure 1.1: Waterfall Model

Requirements: All possible requirements of the system to be developed are recorded and documented in a requirement specification document.

Design: The requirement is studied in this phase and the system design is prepared with logical and physical system specifications. It helps in specifying hardware and system requirements while defining the overall system architecture.

Coding: The units of system is coded and tested in this phase with inputs from the system design. Unit testing is performed for each unit developed.

Verification & Testing: The units developed are integrated into a working system after testing of each unit. After the integration of the system a final testing is done to ensure there is no bugs and errors.

Maintenance: This phase is active when client finds any issues in the system and developers releases better versions to enhance the product.

Following are the reason behind using Waterfall Model as the development methodology:

- In this project necessities, processes and results are well documented, clear and fixed.
- Sufficient resources with essential knowledge are available to support the product.
- There are no ambiguous requirements.
- Easy to arrange tasks.
- All the phases are completed one after another.

1.6 Report Organization

Introduction deals with the introduction of the system with its objectives, problem, scope and limitations along with the reason why the system is made.

Background Study and Literature Review summarizes the work that has been carried out in the field and also describes the features about some existing applications related to the projects by reviewing the existing literature that contains reviews of journals and articles, and earlier thesis related to the subject.

System Analysis and Design focuses on the different requirement of the system, which describes about the functional, non-functional, feasibility analysis. Class, object, sequence, activity, component and deployment diagrams are designed to overview the system and also describes about the implementation of the algorithms used.

Implementation and Testing emphasizes tools used in system development, implementing details and result of test performed.

Conclusion and Future Recommendation highlights brief summary of lesson learnt, outcome and conclusion of the whole project and explain what have been done and what further improvements could be done.

CHAPTER 2

BACKGROUND STUDY AND LITERATURE REVIEW

2.1 Background Study

In recent years, the courier industry in Nepal has been growing rapidly due to the expansion of e-commerce and the increasing demand for fast and reliable delivery services. Many companies in Nepal rely on courier service providers to deliver their products quickly and securely. [3] There are several domestic and international courier service providers operating in Nepal, including KTM Couriers, Kontrakt Express & Cargo Pvt. Ltd, Super Kinetic Nepal Pvt. Ltd and United Nepal Couriers and Cargo Pvt. Ltd. [3] These companies offer a range of services such as door-to-door pickup and delivery, express bulk parcel service, cargo services, and packaging services. [4]

The existing system is a manual one in which users are maintaining ledgers, books etc. to store the information like goods booking details, loading particulars, deliveries particulars, details of receivers of items at all branches, and customer details as well as employee details. It is very difficult to maintain historical data. Also, regular investments need to purchase stationary every year. [5]

The courier service industry in Nepal is facing challenges in adopting new technologies, leading to a lack of modernization. Key issues include the reliance on paper-based billing and invoicing systems, the absence of computerized operations, the lack of a robust package tracking mechanism, and limited delivery status and proof of delivery (POD) systems. These challenges result in inefficiencies, delays, and a lack of transparency and accountability in the delivery process.

To overcome these challenges, courier companies in Nepal need to invest in modernizing their operations. This includes transitioning from paper-based systems to electronic billing and invoicing, implementing computerized systems for order processing and administrative tasks, and establishing a reliable package tracking mechanism. Additionally, there is a need to develop standardized delivery status and POD systems to provide customers with real-time updates and ensure verification of successful deliveries. Training programs and awareness campaigns can also play a vital role in promoting the adoption of new technologies and driving industry-wide modernization efforts.

The vehicle routing problem is a serious and one of the oldest challenges in the logistics and delivery business. George Dantzig and John Ramser first represented it in 1959 in their

paper. In which they wrote the first mathematical approach and later applied it to the petrol deliveries. The goal of this problem is to find the best possible set of routes for the vehicles to reach every stop while minimizing the overall distance traveled, the number of vehicles used, delivery time, and fuel costs. This will also consider other factors, including capacity of each vehicle, the time window, and the service time required per stop. [6]

In the Vehicle Routing Problem (VRP), the goal is to find optimal routes for multiple vehicles visiting a set of locations. A better way to define optimal routes is to minimize the length of the longest single route among all vehicles. This is the right definition if the goal is to complete all deliveries as soon as possible. [7]

2.2 Literature Review

CMS1.0 is a cloud-based courier and cargo management software solution that aims to streamline the end-to-end delivery business processes. It offers a range of features and benefits to improve operational efficiency and customer satisfaction. The system ensures high accuracy by utilizing the latest technologies and minimizing wear and tear. With strict access controls, it prioritizes system safety to protect crucial information. By providing consistent processes, it eliminates human errors and increases reliability. The software enables paperless booking, client/customer management, dispatching, invoice generation, branch management, inventory management, barcode scanning, reporting, customer web portal, online tracking, pickup scheduling, rights management, and easy proof-of-delivery (POD) entry. With these features, CMS 1.0 empowers courier companies to efficiently manage their operations, enhance service quality, and meet customer expectations. [8]

The Courier Management System, developed by Riddha Soft Pvt. Ltd., is a user-friendly solution that simplifies shipment management. Users can easily trace their shipments and receive real-time updates on the status and location of their packages. The system integrates with Sajha accounts, enabling users to add shipments conveniently. It also offers a built-in cost calculation feature, allowing users to estimate shipping costs. Additionally, the system provides delivery time estimations, helping users plan their shipping timelines effectively. With its comprehensive features, the Courier Management System enhances efficiency and transparency in courier operations. [9]

The Courier Management System project allowed users to efficiently manage couriers and parcels. Users could log in as the system admin and perform tasks such as adding and

deleting couriers. The system also facilitated updating the status of couriers as delivered or not. It provided a simple solution for tracking consignments and shipping details, allowing customers to track their orders online. The project included modules for login, order placement, billing, and consignment tracking. Customers and recipients could easily access information about the current status, expected delivery dates, and other relevant details of their consignments. [10]

The Courier Management System using Cloud Computing offers several key functionalities. These include the ability to add new employees to the system, as well as delete or update existing employee information. Additionally, the system allows for the addition of new branches, along with the flexibility to delete or update branch details as needed. In terms of package management, the system enables efficient pickup or receipt of packages, ensuring a smooth workflow. Transit updates provide real-time information on the status and location of packages during transportation. The system also handles the tracking and management of returned packages. Finally, delivery confirmations provide confirmation and verification of successful package delivery. By utilizing cloud computing technology, the Courier Management System offers scalability, accessibility, and data security. It simplifies the management of employees, branches, and packages, ensuring efficient and effective courier operations. [11]

The DHL Courier Management System offers a user-friendly interface for tracking shipment locations, featuring attractive and pleasant interfaces. It ensures efficient operations with effective error handling. The system combines tracking and courier management software into one, and provides a mobile app for both tracking shipments and updating shipment locations, replacing traditional barcode readers. It is compatible with modern operating systems and offers increased efficiency and cost-effectiveness. [12]

CHAPTER 3

SYSTEM ANALYSIS AND DESIGN

3.1 System Analysis

3.1.1 Requirement Analysis

The requirement analysis is a crucial phase in the development of the Courier Management System. It involves analyzing and understanding the collected requirements to identify the system's functionalities, constraints, and scope. Here are the key aspects of the requirement analysis for this project:

Requirement Collection through Internet: Conduct extensive research on the internet to gather additional requirements and insights related to courier management systems. Explore industry best practices, case studies, and customer feedback to identify common features and functionalities that should be considered in the system.

Study of Existing Software: Analyze existing courier management software solutions to understand their features, capabilities, and limitations. Study their user interfaces, functionalities, tracking mechanisms, billing systems, and other relevant aspects. This analysis will provide valuable insights into industry standards and help identify potential areas for improvement in the new system.

Identification of Existing System Limitations: Assess the limitations and shortcomings of the existing courier management systems. Identify the specific areas where they are lacking in terms of functionality, efficiency, customer experience, and overall performance. This analysis will guide the development of a system that addresses these limitations and provides enhanced capabilities.

By conducting comprehensive requirement analysis through internet research and studying existing software, you will gather valuable insights to inform the development of the Courier Management System. This analysis will help identify the specific requirements and functionalities that need to be incorporated into the new system to address the shortcomings of existing systems and meet the needs of stakeholders effectively.

i. Functional Requirement

Different The featured to be included in this project are:

Booking Management: Customers can place book online, providing all necessary details such as sender and recipient information, package dimensions, and delivery preferences. The system performs real-time validation and verification of booking details to ensure accuracy and completeness. Customers and administrators can track the status of orders; from the moment they are placed until the successful delivery. A PDF of booking details is generated after booking is completed.

Package Tracking: The system provides a robust package tracking mechanism that offers real-time updates on the location and status of each package. Packages are assigned unique barcodes that are scanned at different checkpoints, allowing for accurate tracking throughout the delivery process. Automated notifications are sent to customers at various stages, including package pickup, transit, and final delivery.

Delivery Route Optimization: The system utilizes advanced algorithms to optimize delivery routes based on factors such as package volume, delivery time windows, and traffic conditions. By optimizing routes, the system ensures optimal utilization of available resources, such as vehicles and drivers, leading to reduced delivery time and costs.

Customer Communication: Customers can communicate with the courier company through a dedicated interface, allowing them to inquire about order status, request changes, or provide feedback. Customers receive automated notifications regarding order confirmation, estimated delivery time, delays, and successful delivery.

Reporting and Analytics: The system generates comprehensive reports and analytics, providing insights into key performance metrics such as delivery times, resource utilization, and customer satisfaction. The reports are presented in a user-friendly format, enabling administrators to easily analyze data and make informed decisions.

USECASE DIAGRAM

Use Case Diagram for a Courier Management System consist of:

Actors:

Customer: The person or entity that interacts with the CMS to place orders, track packages, and receive notifications.

Admin: The administrator or staff responsible for managing overall system operations.

User: The admin staff responsible for managing orders, packages, delivery agents, and overall system operations.

Delivery Agent: The individual responsible for picking up and delivering packages to customers.

Use Cases:

Add Courier: The Customer can initiate an order by providing sender and recipient details, package information, and delivery preferences.

Check Status: Both the Customer and Admin can track the status and location of a package using a tracking number.

Update Package Status: The Delivery Agent updates the status of a package as it moves through different stages (picked up, in transit, delivered).

Generate Reports: The admin generates reports related to order statistics, delivery performance, and revenue.

Manage Customers: The admin manages customer accounts, including registration, updates, and deletions.

Manage Packages: The admin oversees package details, allocation, and modifications.

Relationships:

Association: Connects actors with use cases to show which actors are involved in each use case.

System Boundary:

The box that encloses all actors and use cases represents the system's boundaries, illustrating what's inside the CMS and what's external to it.

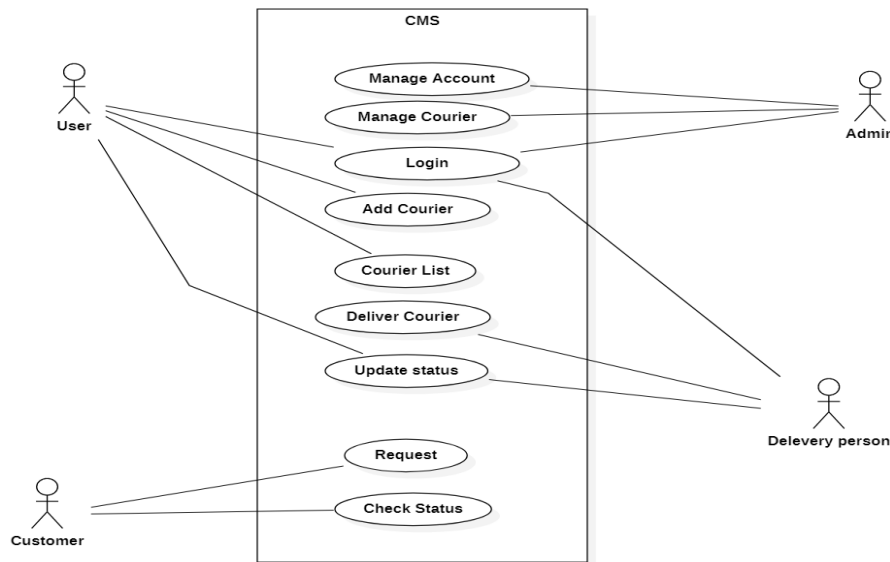


Figure 3.1: Use Case Diagram of Courier Management System

ii. Non-Functional Requirement

Different non-functional requirement has been studied and identified and are listed as below:

Availability: The system will be developed as a web application and can be used online.

Security: The system will be secure and data won't be shared to third party entities. Only authorized entities have access to the data.

Performance: The system will be designed for smooth performance with optimization.

Reliability: The system will be reliable and gives the same result irrespective of the user if same operations are done.

3.1.2 Feasibility Analysis

The feasibility study concluded that the project is able to be implemented to success as it was carefully planned.

- **Technical Feasibility Study:** The system is technically feasible as the requirement for the development of the system is easily accessible. The necessary hardware and software required for the development and implementation of the system is available. The basic programming language which is suitable for project is available and the libraries required for project is capable of achieving the result that we are aiming for. All the existing resources can be used for the development and maintenance system.

- Operational Feasibility Study: The system is easy to operate with the basic knowledge of computer and internet and well-trained manpower is not necessary. User can also easily access the system as it is user friendly in many aspects with good User Interface (UI). This system includes all the requirements used for Courier Management System and this system is completely operational and can be successfully implemented and administration feel easy to use this App as it is user-friendly.
- Economic Feasibility Study: The system is economically feasible and cost effective. As all the tools and resources required are either open sources or free. After the completion of the system organization didn't need to deploy any new hardware and software as the required software and hardware. The existing resource of the system can be used.
- Schedule Feasibility Study: The system is completed within scheduled time and do not exceed the scheduled time.

Table 3.1: Gantt chart Table for Courier Management System

Proposal	20.63 days
Introduction to the Project	2 days
Problem Statement	2 days
Objective	3 days
Methodology	5 days
Project Planning	4 days
Gantt Chart	3 days
Conclusion	1 day
Analysis	26.75 days
Feasibility Analysis	7 days
Requirement Analysis	9 days
Requirement Specification	5 days
Diagrams	5 days
Design	15.5 days
UI Design	8 days
Database Design	7 days
Implementation	29.75 days
Coding	22 days
Development	7 days
Testing	10.5 days
Black-box Testing	4 days
Unit Testing	6 days
Documentation	102.63 days
Reporting	102.63 days

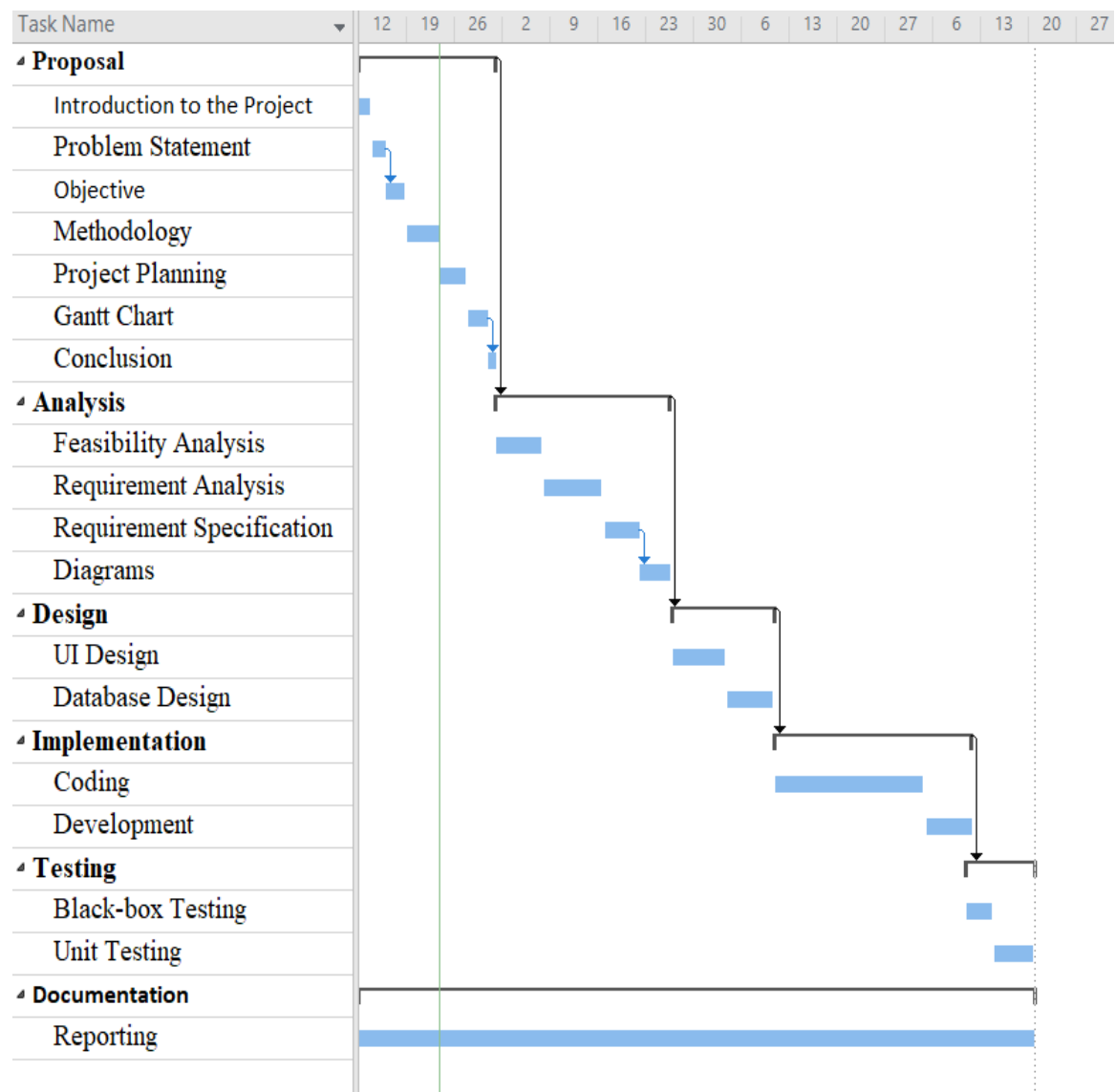


Figure 3.2: Gantt chart for Courier Management System

3.1.3 Object Modelling using Class and Object Diagrams

To create object and class diagrams, it was needed to identify the main objects and their relationships in the context of the chat application. Based on the features described earlier, here's a description of the main objects and their relationships:

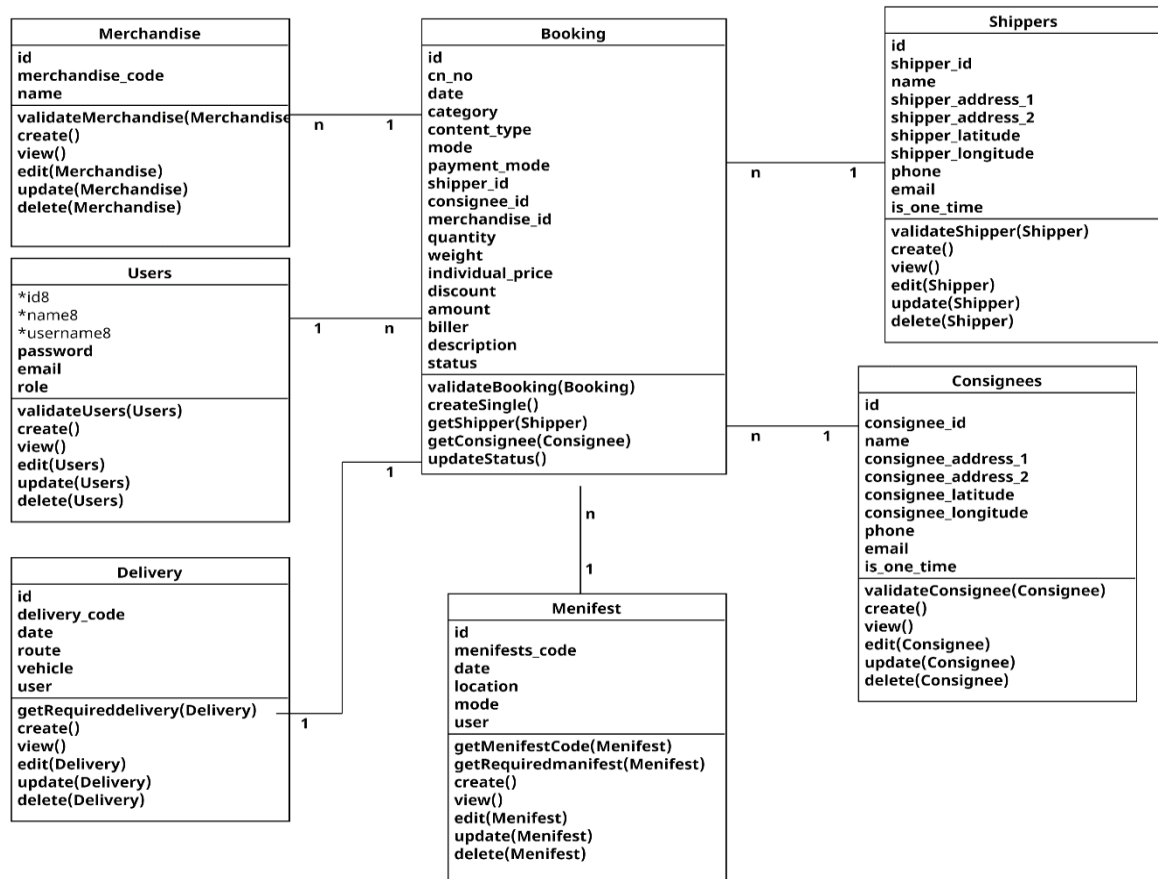


Figure 3.3: Class Diagram for Courier Management System

Figure 3.3 shows Class Diagram of Courier Management System. It shows the communication between different classes.

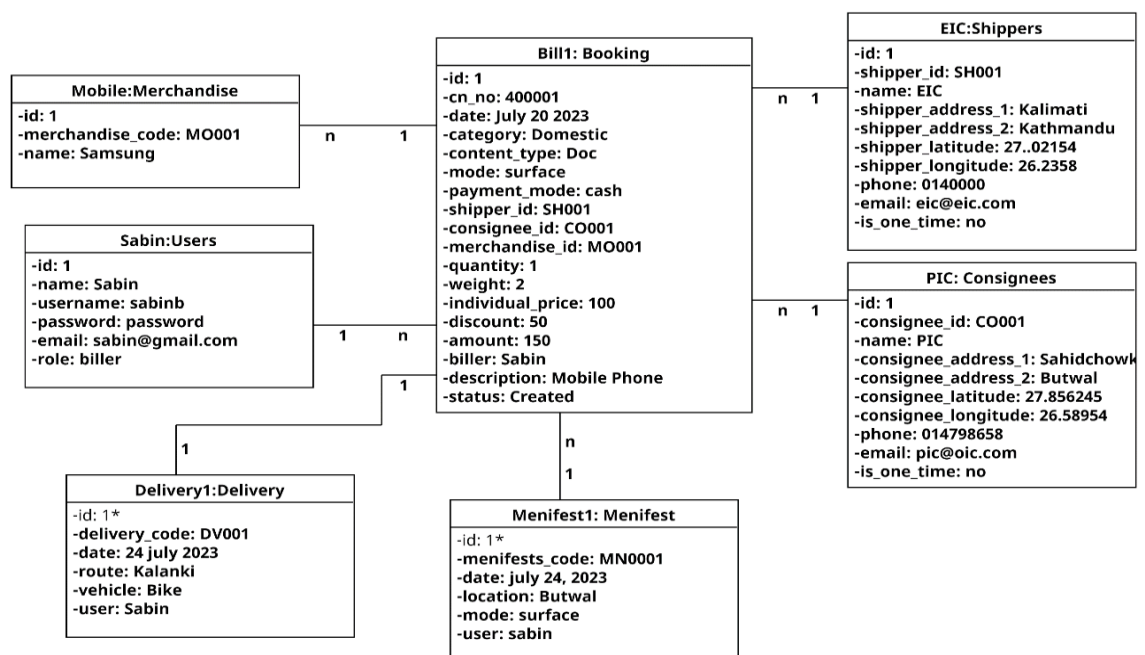


Figure 3.4: Object Diagram for Courier management System

Figure 3.4 shows object Diagram of Courier Management System. It shows the communication between different classes.

3.1.4 Dynamic Modelling using Sequence Diagrams

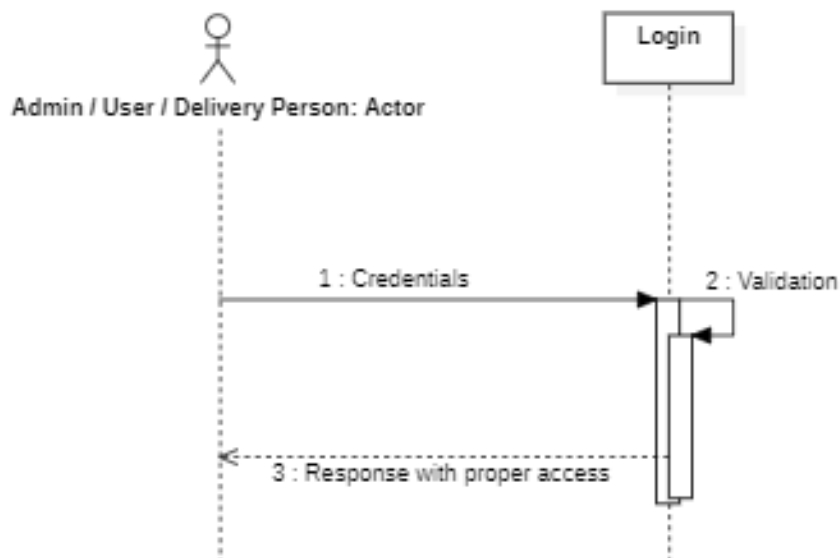


Figure 3.5: Sequence Diagram for login to CMS

Figure 3.5 illustrates how a user login to the system, including validation. It shows the interaction between different users of the system with the system.

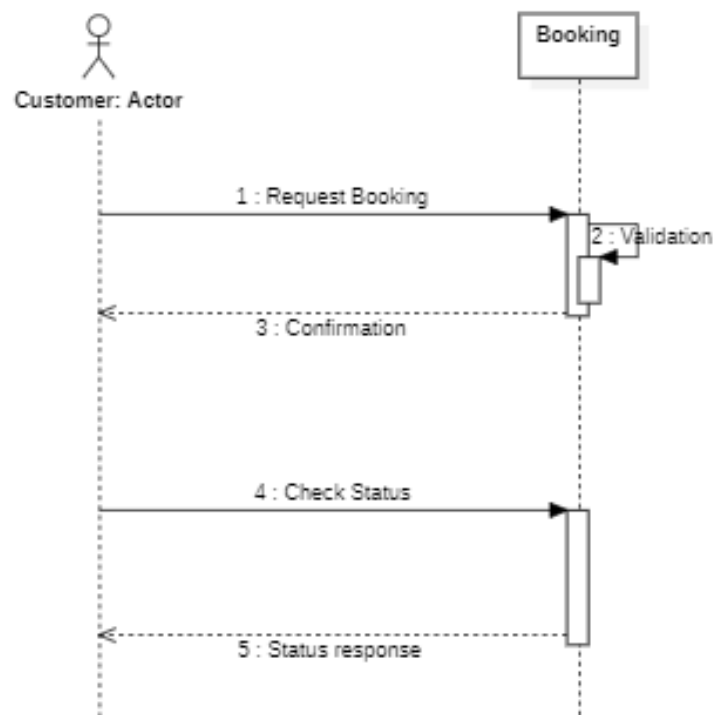


Figure 3.6: Sequence Diagram for booking request in CMS

Figure 3.6 illustrates how a customer's "Request Booking" is processed, including the validation. It also shows how a customer can check the order status.

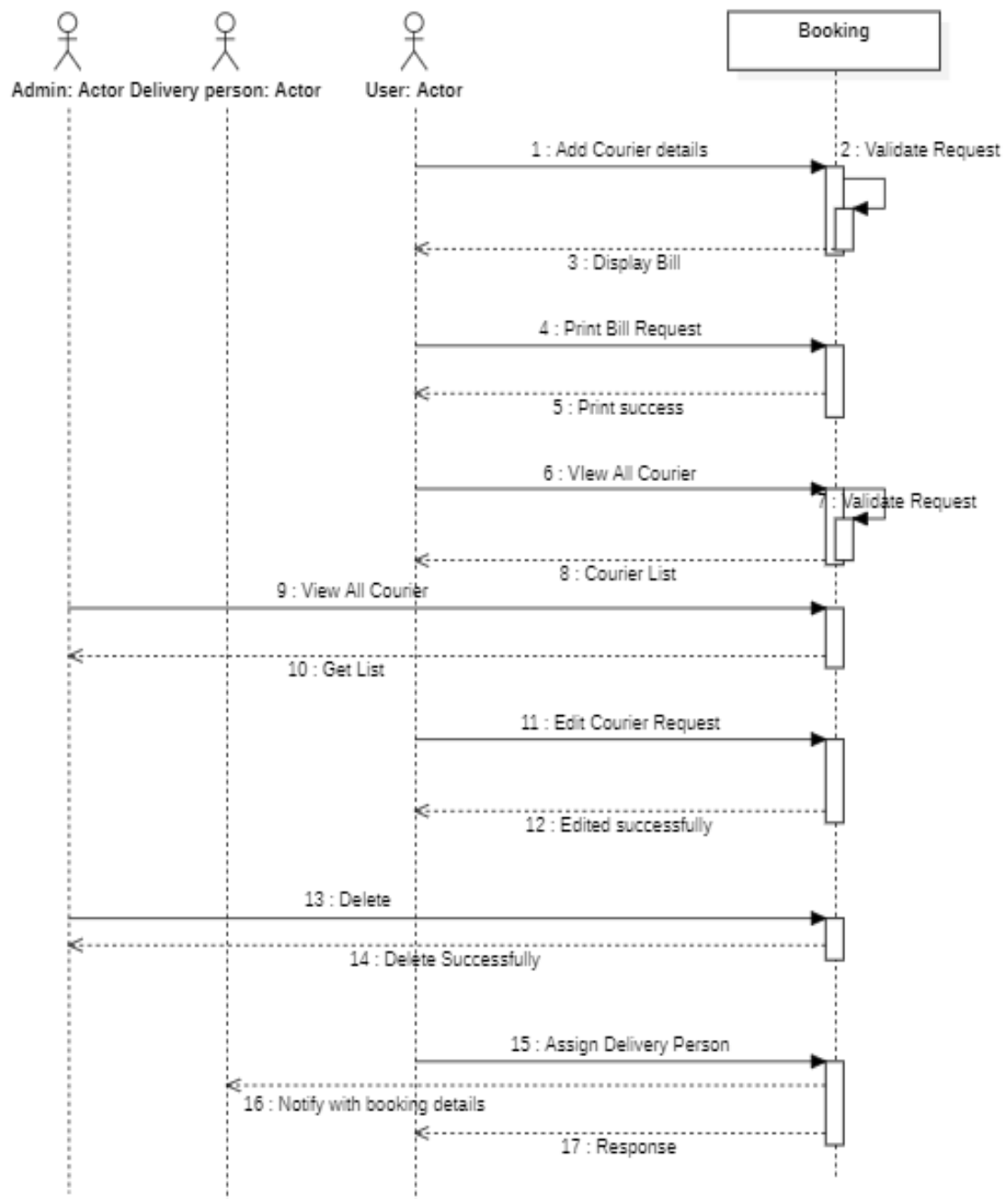


Figure 3.7: Squence Diagram for booking management in CMS

Figure 3.7 illustrates how a user add courier details, including validation, order update, delete and confirmation generation, showing the interactions between the user, admin and the System. It also shows how a delivery person is assigned.

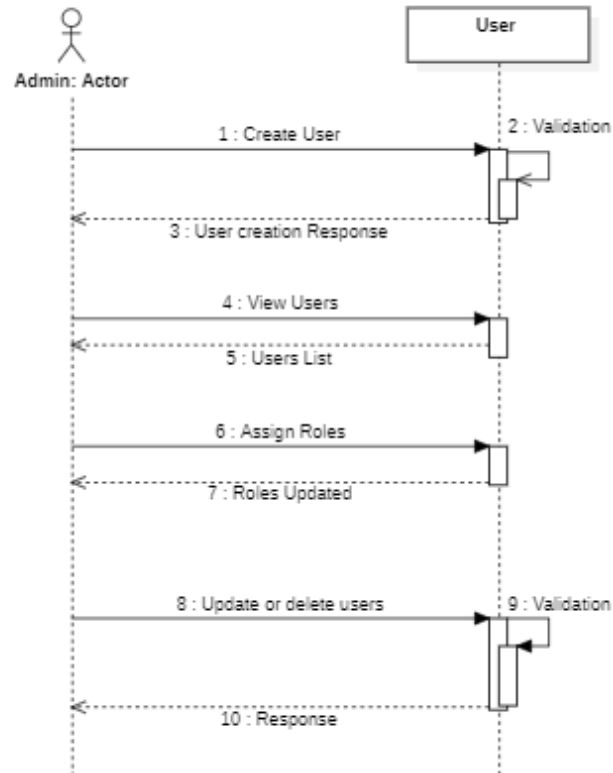


Figure 3.8: Sequence Diagram for user management in CMS

Figure 3.8 demonstrate interactions between the Admin and System, this diagram outlines how user information is accessed, updated, and validated within the system for effective user management.

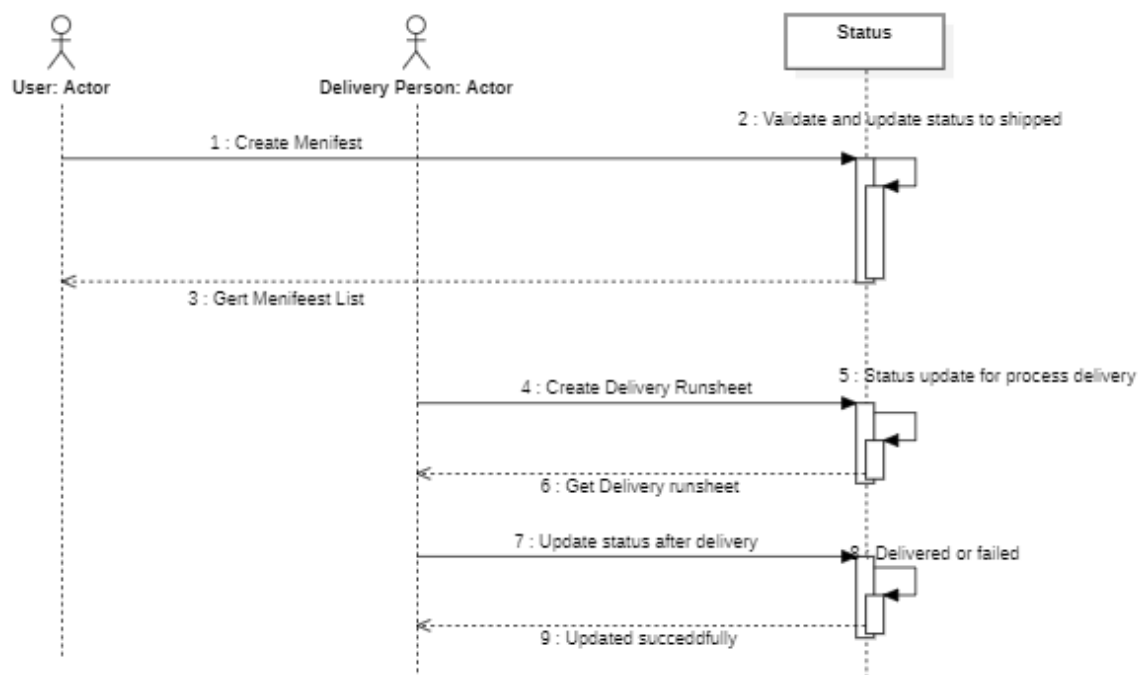


Figure 3.9: Sequence Diagram for booking status in CMS

Figure 3.9 demonstrate interactions between the User, Delivery Person and System, this diagram outlines how delivery status is updated by the system user and the delivery person.

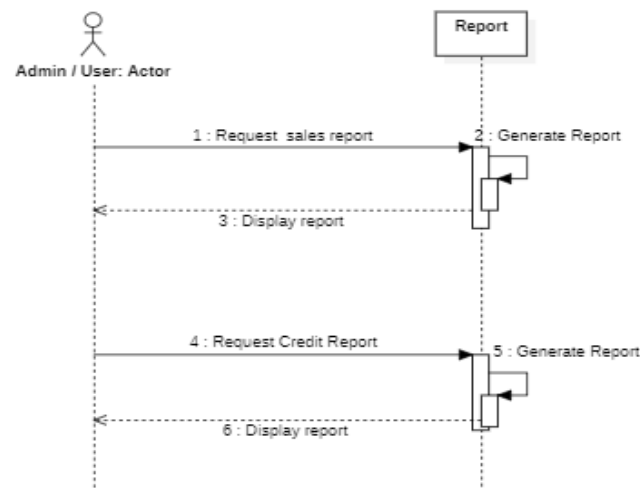


Figure 3.10: Sequence Diagram for report management in CMS

Figure 3.10 demonstrate interactions between the User, Admin and System, this diagram outlines how different reports are generated in the system.

3.1.5 Process Modelling using Activity Diagrams

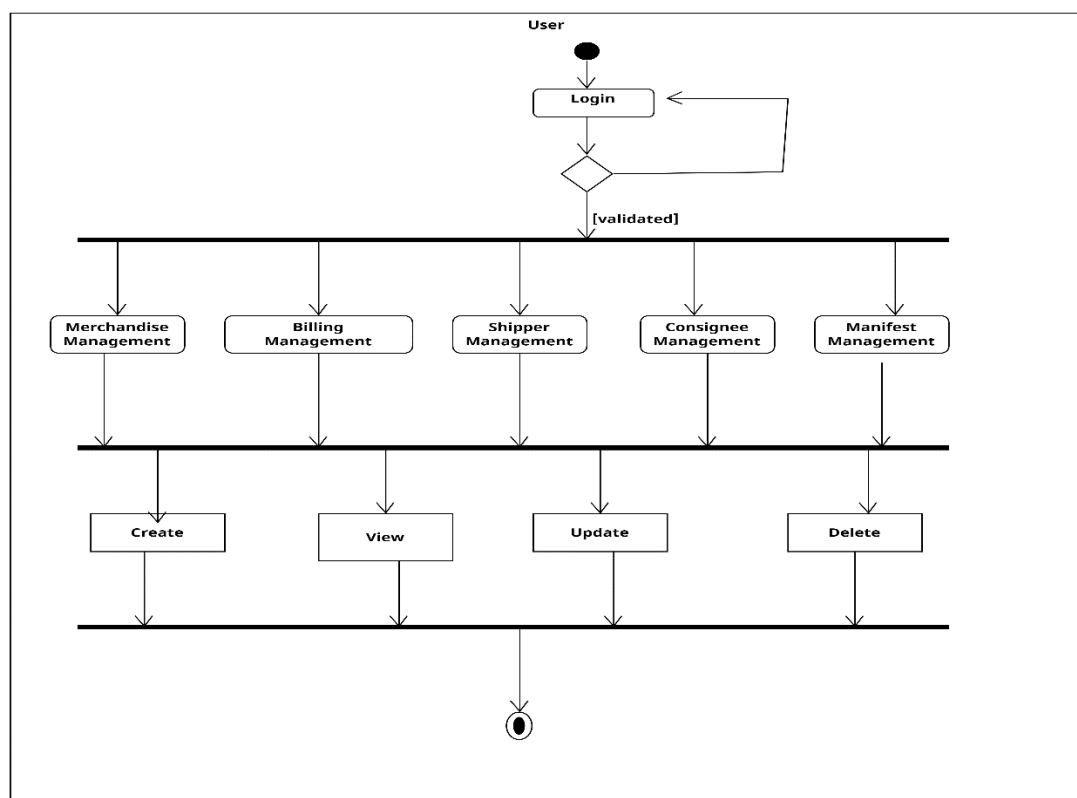


Figure 3.11: User Activity Diagram for Courier Management System

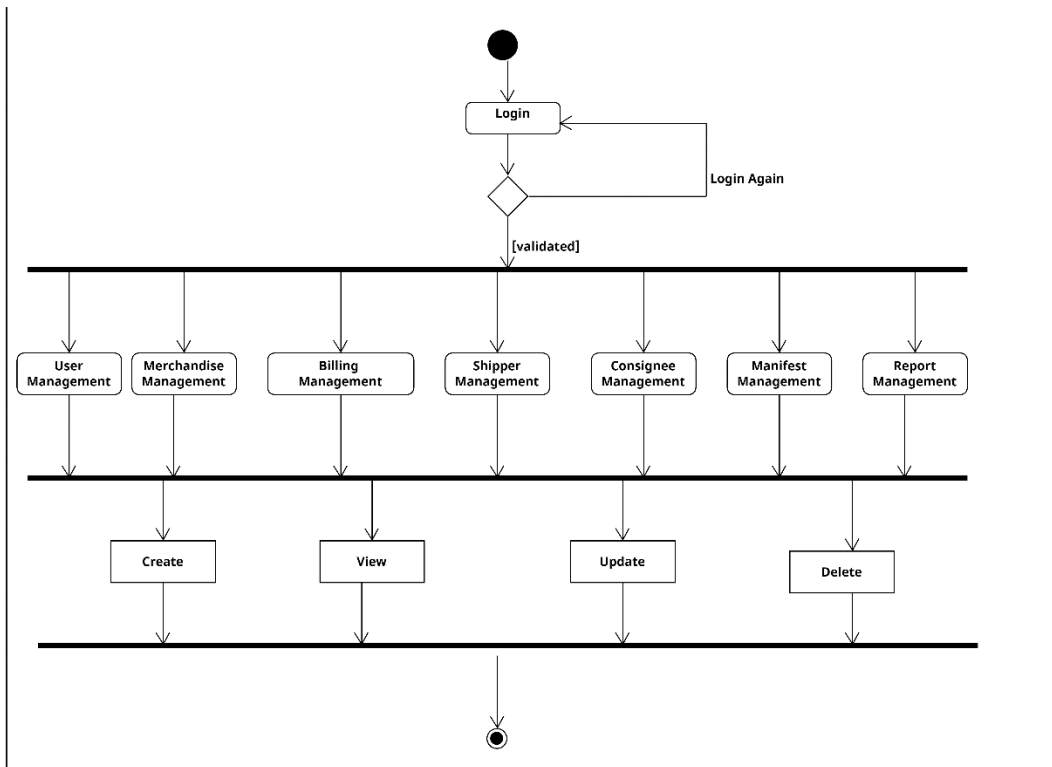


Figure 3.12: Admin Activity Diagram for Courier Management System

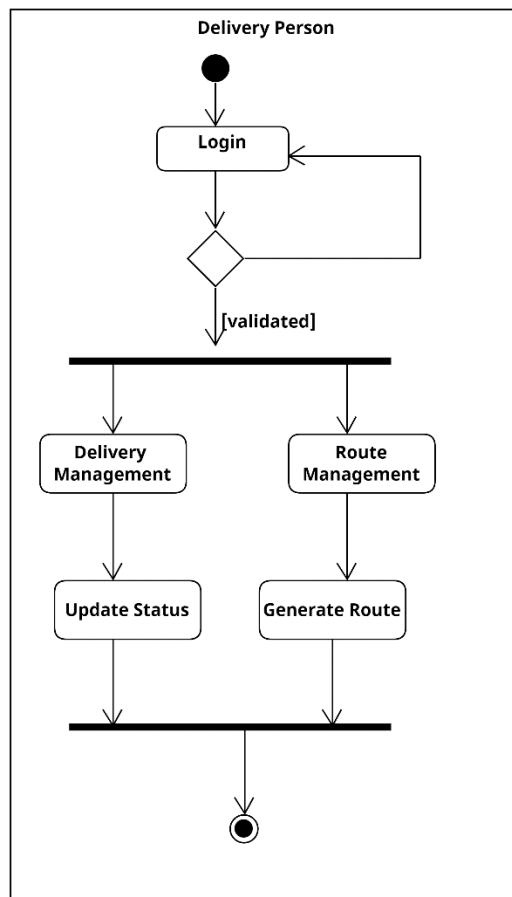


Figure 3.13: Delivery person activity diagram for Courier Management System

3.2 System Design

To realize the different functional requirement of the system in graphical form, different design diagram of the system has been prepared which are as follows:

3.2.1. Refinement of Class diagram

Refinement of class diagram involves adding more details and clarifying relationships between classes. It is a process of elaborating on the initial representations to provide a more comprehensive view of the system. Let's refine the previously provided class and object diagrams for the system:

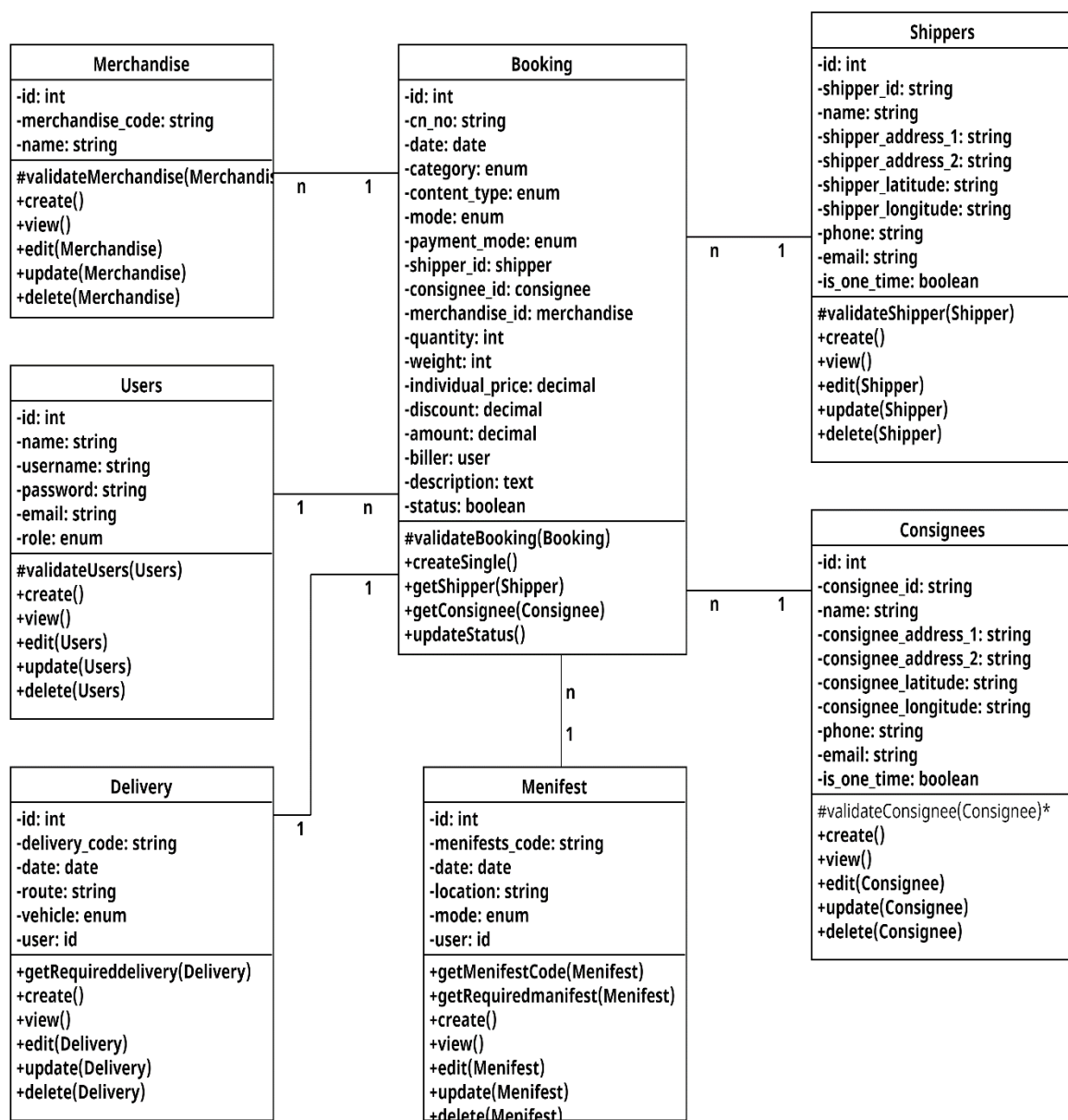


Figure 3.14: Refinement of class diagram for Courier Management System

3.2.2. Component Diagrams

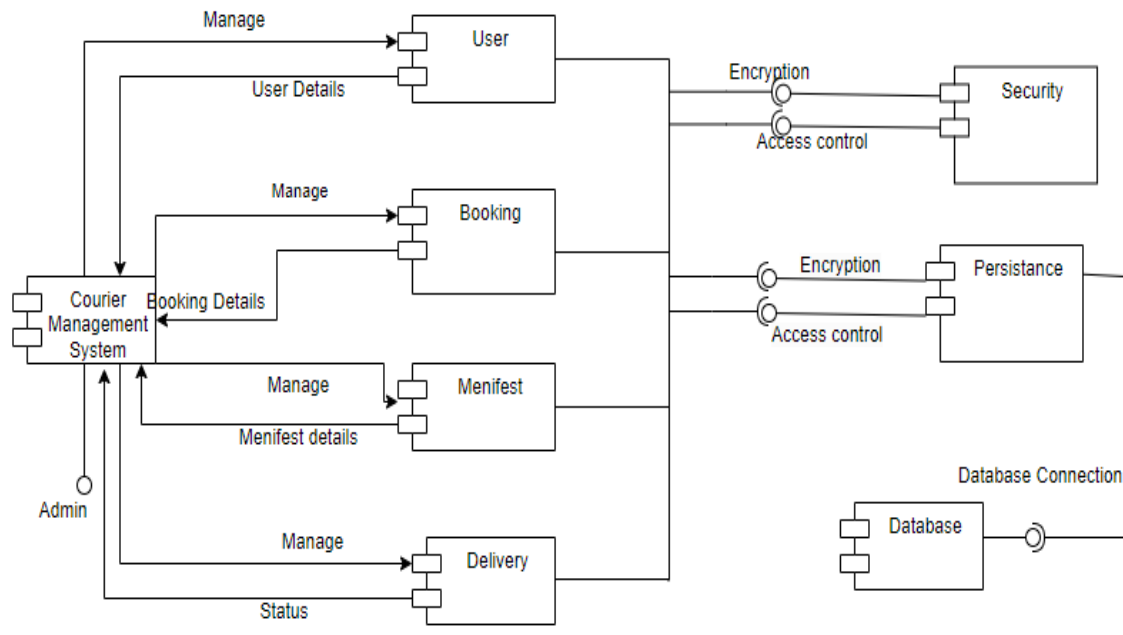


Figure 3.15: Component Diagram of Admin for Courier Management System

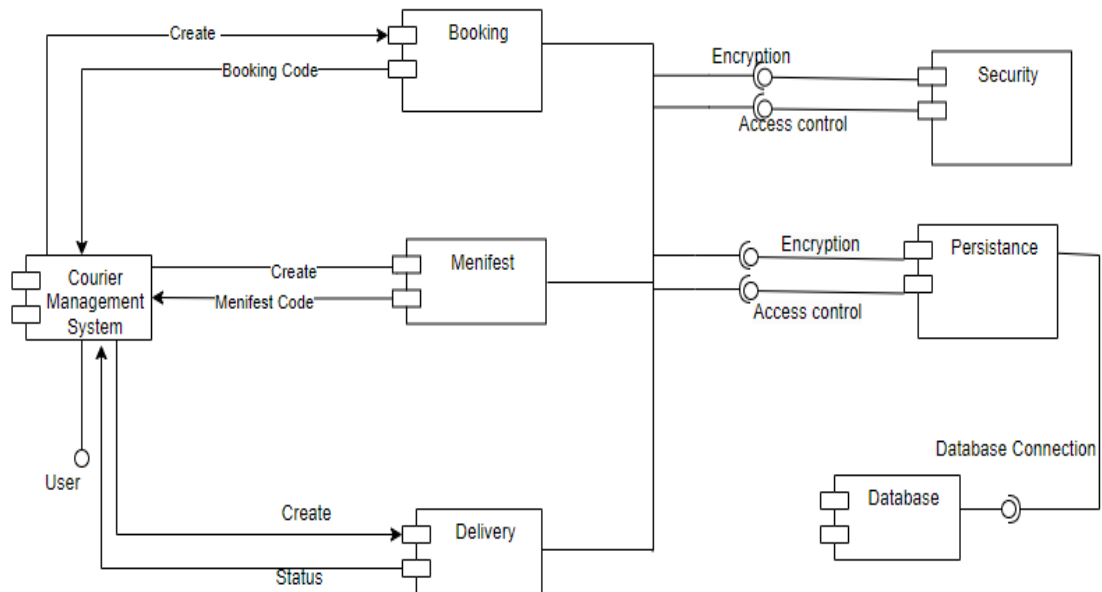


Figure 3.16: Component Diagram of Admin for Courier Management System

3.2.3. Component Diagrams

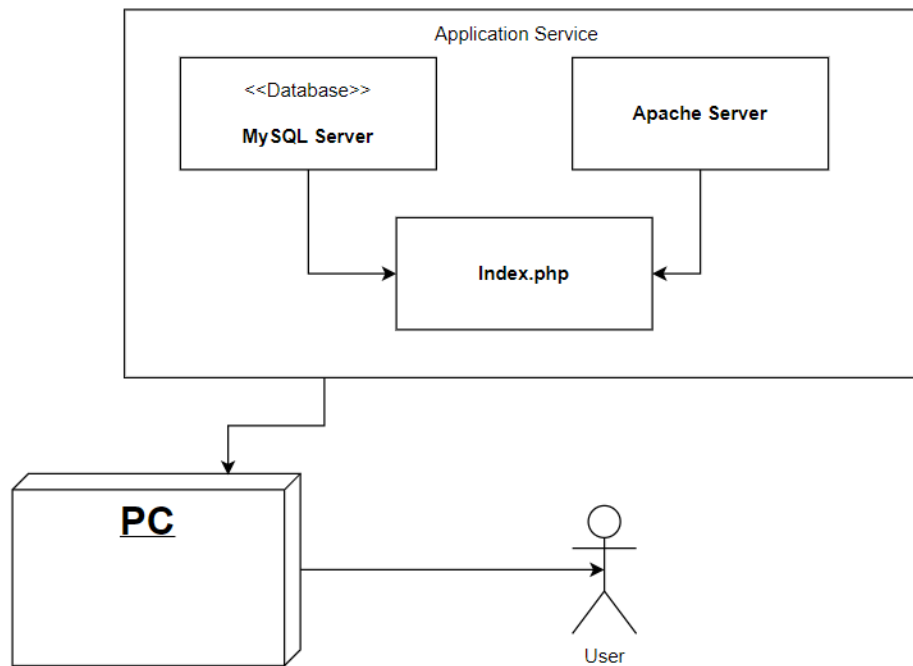


Figure 3.17: Deployment Diagram for Courier Management System

3.3 Algorithm Details

3.3.1. Vehicle Routing Problem Algorithm

This Vehicle Routing Problem (VRP) uses the nearest neighbor heuristic. The VRP is a combinatorial optimization problem in which a fleet of vehicles is tasked with delivering goods to a set of customers while minimizing the total travel distance or cost.

Haversine Distance Calculation: This function calculates the distance between two geographic coordinates (latitude and longitude) using the Haversine formula. This distance represents the distance between two points on the Earth's surface, considering the curvature of the Earth.

$$a = \sin^2(\Delta\text{lat}/2) + \cos(\text{lat1}) * \cos(\text{lat2}) * \sin^2(\Delta\text{long}/2)$$

$$c = 2 * \text{atan2}(\sqrt{a}, \sqrt{1-a})$$

$$d = R * c$$

Where:

Δlat is the difference in latitude between the two points.

Δlong is the difference in longitude between the two points.

lat1 and lat2 are the latitudes of the two points.

R is the radius of the Earth (mean radius = 6,371 km).

d is the distance between the two points.

Distance Matrix Generation: This function creates a distance matrix that contains the pairwise distances between all customer locations and the depot (starting and ending point). It uses the Haversine distance calculation to compute these distances.

Nearest Neighbor Heuristic: This function is the core of the VRP algorithm. It uses the nearest neighbor heuristic to construct routes for the vehicles. The nearest neighbor heuristic starts from the depot and repeatedly selects the nearest unvisited customer to the current customer until the capacity of the vehicle is exceeded or no unvisited customers remain. This process is repeated until all customers are served.

Demand for Customers: A constant demand value is assigned to each customer, indicating the quantity that needs to be delivered to that customer. This demand is assumed to be the same for all customers for this project.

This outputs the routes for each route option, including the depot, customers in the order they are visited, and the depot again. It also calculates and displays the total distance traveled for each route option.

3.3.2. Algorithm for calculating price based on distance

One commonly used algorithm for this purpose is the Haversine formula. The Haversine formula calculates the distance between two points on the surface of a sphere (such as the Earth) given their longitudes and latitudes. This formula takes into account the curvature of the Earth and provides accurate distance calculations.

The Haversine can be expressed in trigonometric function as:

$$a = \sin^2(\Delta\text{lat}/2) + \cos(\text{lat1}) * \cos(\text{lat2}) * \sin^2(\Delta\text{long}/2)$$

$$c = 2 * \text{atan2}(\sqrt{a}, \sqrt{1-a})$$

$$d = R * c$$

Where:

Δlat is the difference in latitude between the two points.

Δlong is the difference in longitude between the two points.

lat1 and lat2 are the latitudes of the two points.

R is the radius of the Earth (mean radius = 6,371 km).

d is the distance between the two points.

The working of this algorithm can be described as,

- Defining the price zones such as different prices for different zones. The zones are categorized based on the distance between the places.
- Calculating the distance using Haversine Formula.
- Assigning the zone and calculating the price logics. The price logic has pre-defined base price, base distance and additional rate on every increase in 5 Km.

$\text{Price} = \text{base_price} + (\text{distance} - \text{base_distance}) * \text{additional price per 5 Km.}$

3.3.3. Algorithm for scanning the QR code from the uploaded image

Image Preprocessing: It starts by preprocessing the input image to enhance its quality and extract necessary features. This might involve operations like grayscale conversion, noise reduction, and contrast adjustment to improve barcode visibility.

Feature Detection: It uses feature detection techniques to locate potential barcodes regions in the image. These features could be edges, corners, or other distinct patterns that characterize barcodes regions.

Region of Interest (ROI) Detection: Once potential barcode regions are identified, this algorithm to determine regions of interest (ROIs) that are likely to contain barcodes. These ROIs are then further analyzed for barcode content.

Decoding and Recognition: The algorithm decode and recognize the barcode content. The algorithms analyze the patterns and structures within the barcode, including the positioning of bars and spaces.

CHAPTER 4

IMPLEMENTATION AND TESTING

4.1 Implementation

4.1.1. Tools Used (CASE tools, Programming language, Database platforms)

Following are the tools and framework used for the accomplishment of this project:

Front End Tools

- **HTML**
HTML is used to design forms, table and different pages with page elements such as header, footer, dashboard etc.
- **CSS**
CSS is used to support the design by styling the different components in the projects. It is used to update the feel and look of the project.
- **JavaScript**
In this project, JavaScript is used to make the project more responsive and user friendly.
- **Tailwind**
The tailwind is used to style the elements of HTML such as forms, tables, and different tags such as div and spans.

Back End Tools

- **PHP & Laravel**
PHP and its framework Laravel are used as server-side language in this project to perform the CRUD operation in database along with maintaining session during browsing the project.

Server

- **XAMPP SERVER**
XAMPP is used to run Apache server for PHP and MYSQL server for the database in this project development.

Database

- **MySQL**
MYSQL database is used as the database for the project to store different data of the projects.

Documentation Tools

- MS Office

This is used for writing and editing the documentation of Courier Management System.

- StarUML

This is used to generate diagrams for system analysis and design of Courier Management System. Diagrams were created using this tool in order to save time since all components are available with drag and drop functions.

- MS Project

MS project is used to design the time frame of the project using Gantt chart function on it.

4.1.2. Implementation Details of Modules (Description of procedures / Functions / Methods)

4.2 Testing

System testing is done by giving different training and testing datasets. This test is done to evaluate whether the system is providing accurate summary or not. During the phase of the development of the system, our system is tested time and again. The series of testing conducted are as follow:

4.2.1. Test Cases for Unit Testing

In unit testing, we designed the entire system in modularized pattern and each module is tested. Until we get the accurate output from the individual module, we work on the same module. The input forms are tested so that they do not accept invalid input.

User Login

Table 4.1: Test case for User Login of Courier Management System

S. N	Test Name	Input	Expected Outcome	Actual Outcome	Test Result
1	Enter invalid username & password.	U = Raja Ram Bhurtel P = Test@123	Login fail due to wrong username.	Login Failed.	Pass

2	Enter valid username, password and click login button.	Username = Raja Ram Bhurtel Password = Test@123	Login Successful and redirect to dashboard page.	Redirect to dashboard page.	Pass
---	--	--	--	-----------------------------	------

Booking

Table 4.2: Test case for Booking of Courier Management System

S. No	Test Name	Input	Expected Outcome	Actual Outcome	Test Result
1	Enter the booking details.	Shipment details with consignee and sender details.	Booking saved and redirected to POD page.	Booking Saved.	Pass

Create Manifest

Table 4.3: Test case for Create Manifest of Courier Management System

S. No	Test Name	Input	Expected Outcome	Actual Outcome	Test Result
1	Scan the Image.	Booking Code.	Load The details after scan.	Details popup in the input field.	Pass

Create Delivery

Table 4.4: Test case for Create Delivery of Courier Management System

S. No	Test Name	Input	Expected Outcome	Actual Outcome	Test Result
1	Scan the Image.	Booking Code.	Load The details after scan.	Details popup in the input field.	Pass

Create User

Table 4.5: Test case for Create User of Courier Management System

S. No	Test Name	Input	Expected Outcome	Actual Outcome	Test Result
1	Enter User details.	Name: Sabin Email:sabinbhurtel@gmail.com Pass: test@1	User Created Message.	No message.	Failed.
2	Enter User details.	Name: Sabin Email:sabinbhurtel@gmail.com Pass: test@1	User Created Message.	User Created Message Popup.	Pass

Create Merchandise

Table 4.6: Test case for Create Merchandise of Courier Management System

S. No	Test Name	Input	Expected Outcome	Actual Outcome	Test Result
1	Enter Merchandise details.	Code: 001 Name: Book	Merchandise Created Message.	No message.	Failed.
2	Enter Merchandise details.	Code: 001 Name: Book	Merchandise Created Message.	Merchandise Created Message Popup.	Pass

Create Consignee

Table 4.7: Test case for Create Consignee of Courier Management System

S. No	Test Name	Input	Expected Outcome	Actual Outcome	Test Result
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1	Enter Consignee details.	Name: Sabin Address: Thankot	Consignee Created Message.	No message.	Failed.
2	Enter Consignee details.	Name: Sabin Address: Thankot	Consignee Created Message.	Consignee Created Message Popup.	Pass

Create Shipper

Table 4.8: Test case for Create Shipper of Courier Management System

S. No	Test Name	Input	Expected Outcome	Actual Outcome	Test Result
1	Enter Shipper details.	Name: Sabin Address: Pokhara	Shipper Created Message.	No message.	Failed.
2	Enter Shipper details.	Name: Sabin Address: Pokhara	Shipper Created Message.	Shipper Created Message Popup.	Pass

Check Status

Table 4.9: Test case for Check Status of Courier Management System

S. No	Test Name	Input	Expected Outcome	Actual Outcome	Test Result
1	Enter Shipment Code.	Code: 400001	Shipment detail box.	Shipment detail box popup.	Pass.

Report Generation

Table 4.10: Test case for Report Generation of Courier Management System

S. No	Test Name	Input	Expected Outcome	Actual Outcome	Test Result
1	Enter Date range.	2023-03-04 to 2023-03-10	Report of that period should be shown.	Report Generated.	Pass.

4.2.2. Test Cases for System Testing

In system testing, whole system is tested as below:

Table 4.11: System test case for Courier Management System

S. No	Test Name	Input	Expected Outcome	Actual Outcome	Test Result
1	Login to the system with just created credentials.	Username = raja Password = raja@2018	Login Successful and redirect to dashboard.	Redirect to dashboard.	Pass.
2	Enter the booking details.	Booking Details.	Booking save and redirected to POD page.	Booking Saved.	Pass.
3	Scan Image for manifest.	Booking Code from image scan.	Booking details should popup.	Booking Details popup	Pass.
4	Scan Image for Delivery.	Add booking code.	Booking details should popup.	Booking Details popup and delivery saved.	Pass.
2	Logout from the system.	Click the logout button.	Logs out of the system.	User is logged out and redirected to login page.	Pass.

4.3 Result Analysis

Distance based price calculation

For distance-based price calculation, the dataset is taken as longitude and latitude between two locations and after supplying quantity and weight we will get the result. The objective is to predict price based on the distance. The dataset is preprocessed by removing unnecessary data of dataset

Table 4.12: Dataset parameters for distance-based price calculation

S. No	Attributes
1	Latitude_from
2	Longitude_from
3	Latitude_to
4	Longitude_to
5	Weight
6	Quantity

Accuracy Measurement

Different accuracy measurement techniques like precision, recall and F-score were used.

Accuracy Measures are as follows:

$$\text{Precision} = \frac{\text{True Positive}}{\text{True Positive} + \text{False Positive}}$$

$$\text{Recall} = \frac{\text{True Positive}}{\text{True Positive} + \text{False Negative}}$$

$$F - score = \frac{2 * Precision * Recall}{Precision + Recall}$$

For this, we collected the datasets from different sources which consist the data as coordinates and actual distance. After computing distance from haversine formula we got the relative distance. Then, we calculated the absolute difference between actual distance and relative distance. If this absolute value falls below the threshold value, then it is true positive else if relative distance is less than or equals to actual distance, then it is false negative. If both conditions don't satisfy then its false positive.

Table 4.13: Obtained Result

Attributes	Result
True Positive	15
False Positive	2
False Negative	0

$$Precision = \frac{15}{15 + 2} = 0.8823$$

$$Recall = \frac{15}{15 + 0} = 1$$

$$F - score = \frac{2 * 0.8823 * 1}{0.8823 + 1} = 0.9375$$

Vehicle Routing Problem

For vehicle routing problem, the dataset is taken as longitude and latitude between multiple places.

Table 4.14: Dataset for Vehicle Routing Problem

S. No	Attributes
1	Latitude
2	Longitude
3	Vehicle Capacity

Model Building

Step 1: Define the haversine function

```
function haversineDistance($lat1, $lon1, $lat2, $lon2) {
    $lat1 = deg2rad($lat1);
    $lon1 = deg2rad($lon1);
    $lat2 = deg2rad($lat2);
    $lon2 = deg2rad($lon2);

    $earthRadius = 6371;

    $dlat = $lat2 - $lat1;
    $dlon = $lon2 - $lon1;
    $a = sin($dlat / 2) ** 2 + cos($lat1) * cos($lat2) * sin($dlon / 2) ** 2;
    $c = 2 * atan2(sqrt($a), sqrt(1 - $a));
    $distance = $earthRadius * $c;

    return $distance;
}
```

Step 2: Build the distance matrix

```
function buildDistanceMatrix($coordinates) {
    $numPoints = count($coordinates);
    $distanceMatrix = [];

    for ($i = 0; $i < $numPoints; $i++) {
        $distanceMatrix[$i] = [];

        for ($j = 0; $j < $numPoints; $j++) {
            if ($i == $j) {
                $distanceMatrix[$i][$j] = 0; // Distance to itself is 0
            } else {
                $lat1 = $coordinates[$i]["lat"];
                $lon1 = $coordinates[$i]["lon"];
                $lat2 = $coordinates[$j]["lat"];
                $lon2 = $coordinates[$j]["lon"];

                $distance = haversineDistance($lat1, $lon1, $lat2, $lon2);
                $distanceMatrix[$i][$j] = $distance;
            }
        }
    }

    return $distanceMatrix;
}
```

Step 3: Define function for nearest customer

```
function findNearestCustomer($currentCustomer, $visited, $distanceMatrix) {
    $minDistance = PHP_INT_MAX;
    $nearestCustomer = -1;
    $numCustomers = count($visited);

    for ($i = 0; $i < $numCustomers; $i++) {
        if (!$visited[$i] && $distanceMatrix[$currentCustomer][$i] < $minDistance) {
            $minDistance = $distanceMatrix[$currentCustomer][$i];
            $nearestCustomer = $i;
        }
    }

    return $nearestCustomer;
}
```

Step 4: Find the nearest neighbor

```
function nearestNeighbor($distanceMatrix, $vehicleCapacity) {
    $numCustomers = count($distanceMatrix) - 1;
    $visited = array_fill(0, $numCustomers, false);
    $routes = [];

    $currentCustomer = 0;

    $totalDistance = 0;
    $currentRoute = [];
    $currentLoad = 0;

    while (in_array(false, $visited)) {
        $nearestCustomer = findNearestCustomer($currentCustomer, $visited, $distanceMatrix);

        if ($nearestCustomer == -1) {
            break;
        }

        if ($currentLoad + 1 <= $vehicleCapacity) {
            $currentRoute[] = $nearestCustomer;
            $visited[$nearestCustomer] = true;
            $currentLoad += 1;
            $totalDistance += $distanceMatrix[$currentCustomer][$nearestCustomer];
            $currentCustomer = $nearestCustomer;
        } else {
            $currentRoute[] = 0;
            $routes[] = $currentRoute;
            $currentRoute = [0];
            $currentLoad = 0;
            $totalDistance += $distanceMatrix[$currentCustomer][0];
            $currentCustomer = 0;
        }
    }

    if (!empty($currentRoute)) {
        $currentRoute[] = 0;
        $routes[] = $currentRoute;
    }

    return ["routes" => $routes, "totalDistance" => $totalDistance];
}
```

Step 5: Processing data

```
$vehicleCapacity = 5;
$jsonData = '[
  {"location": "Office", "lat": 27.69767654394376, "lon": 85.29203955659513},
  {"location": "Kalanki", "lat": 27.69331368823341, "lon": 85.28168016418789},
  {"location": "Gurjudhara", "lat": 27.688658328533737, "lon": 85.24245789945164},
  {"location": "Nagdhunga", "lat": 27.705524077700307, "lon": 85.208280408543},
  {"location": "Swayambhu", "lat": 27.717031557421695, "lon": 85.28370503208029},
  {"location": "Balaju", "lat": 27.727275304186158, "lon": 85.30469868807273},
  {"location": "Goldhunga", "lat": 27.75943257553317, "lon": 85.28555741615676},
  {"location": "Basundhara", "lat": 27.742394921885253, "lon": 85.3319729180005}
]';
$coordinates = json_decode($jsonData, true);

$result = nearestNeighbor(buildDistanceMatrix($coordinates), $vehicleCapacity);
$routes = $result["routes"];
$totalDistance = $result["totalDistance"];

foreach ($routes as $index => $route) {
  echo "Route " . ($index + 1) . ": " . implode(" -> ", array_map(function ($customer) use ($coordinates) {
    return $coordinates[$customer]["location"];
  }, $route)) . " <br>";
}

echo "Total Distance: " . $totalDistance . " kilometers";
```

Defining different vehicle capacity, we got different result as:

Capacity: 2

```
Route 1: Office -> Kalanki -> Office
Route 2: Office -> Swayambhu -> Balaju -> Office
Route 3: Office -> Gurjudhara -> Nagdhunga -> Office
Route 4: Office -> Goldhunga -> Office
Total Distance: 34.465459081758 kilometers
```

Capacity: 4

```
Route 1: Office -> Kalanki -> Swayambhu -> Balaju -> Office
Route 2: Office -> Gurjudhara -> Nagdhunga -> Goldhunga -> Office
Total Distance: 28.172925576297 kilometers
```

Capacity: 10

```
Route 1: Office -> Kalanki -> Swayambhu -> Balaju -> Goldhunga -> Gurjudhara -> Nagdhunga -> Office
Total Distance: 22.967968322707 kilometers
```

In this model, we first calculate the distance matrix using haversine formula with the given dataset. After getting the distance matrix we use the matrix to find the nearest customer and nearest neighbor. The nearest neighbor function returns us the route after sorting them. Hence, from this VRP we studied that increasing the vehicle capacity we can decrease the total distance covered. And the more the package is from same location it will be faster to deliver those in time.

CHAPTER 5

CONCLUSION AND FUTURE RECOMMENDATIONS

5.1 Lesson Learnt / Outcome

Every project makes us to learn and gain the knowledge in different aspects. In the following project, we have learned lots of problem-solving skills and learn things like team work, finding the solution on our own, proper use of guidelines, communication and writing skills and management of team.

Problem Solving Skills: The project provided valuable opportunities to enhance problem-solving skills. It involved identifying and resolving various issues and errors that occurred within the system. This experience improved our ability to troubleshoot and find effective solutions.

Writing Skills: The project taught us the importance of effective communication through written documentation. We gained proficiency in preparing proposals and project-related documentation. Additionally, we learned to utilize various case tools for creating diagrams such as use case diagrams, schema diagrams, data flow diagrams, activity diagrams, component diagrams, object and class diagrams, and deployment diagrams.

Time Management: Managing time efficiently was a critical lesson learned from this project. We realized the significance of prioritizing project components based on their complexity. This skill allowed us to allocate our time and resources effectively, ensuring that the project progressed smoothly and met its deadlines.

Independence and Self-Reliance: The project encouraged us to become more independent problem solvers. We learned how to tackle challenges and find solutions on our own, which fostered self-reliance and resourcefulness.

Project Management: Managing the courier management system project gave us insights into project management principles. We learned how to plan, execute, and monitor the project's progress, ensuring that it remained on track and within scope.

Hence, the Courier Management System Project provided us with a holistic learning experience, encompassing technical skills, time management, and effective

communication. These lessons will be invaluable for future endeavors in the field of software development and project management.

5.2. Conclusion

In conclusion, the Courier Management System project has successfully achieved its objectives by implementing a range of features and functionalities. The work completed during this project includes the development of efficient booking management, streamlined delivery with route optimization using the Vehicle Routing Problem (VRP) algorithm, comprehensive user management, and effective consignee and shipper management, along with a packet tracking system.

This project had two primary goals. Firstly, it aimed to elevate the efficiency of order processing and substantially minimize delivery durations through the implementation of intelligent route optimization. Utilizing the VRP algorithm, the system identifies and employs the most efficient delivery routes, leading to swifter and more economical deliveries. Secondly, the project aimed to enhance the overall customer experience by incorporating a self-service portal. This portal empowers customers to autonomously track their shipments, offering up-to-the-minute details regarding the status and whereabouts of their packages. This valuable feature not only boosts customer satisfaction but also alleviates the workload on customer support services.

Through the implementation of these features, the Courier Management System has effectively addressed several key problems in the courier industry. Manual order processing has been replaced with automated systems, leading to greater accuracy and speed. Inefficient package tracking has been eliminated, ensuring that customers have full visibility into the progress of their shipments. Lastly, ineffective delivery route planning is now a thing of the past, as the VRP algorithm optimizes routes for maximum efficiency.

Hence, the Courier Management System has successfully tackled industry challenges and delivered a comprehensive solution that benefits both the courier company and its customers. This project demonstrates the potential for technology-driven improvements in logistics and delivery services.

5.3 Future Recommendations

The development project could have been more efficiently handled with regards to design and development. The documentation process might have been better programming the project prior to any documentation

- Advance functions such as multiple route planner.
- Customer communication through chat feature.
- More features like add new branch, organization.
- More users Friendly.
- Remove bug from the system
- More security feature implementation.

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APPENDICES: SYSTEM SCREENSHOTS

LOGIN

Username or Email

Password

LOGIN

[Forgot password?](#)

COURIER
Management System

Dashboard

Status

Guest

Check Status

Home » Check Status

Check Status by CN Number

CONSIGNEE	DESTINATION	STATUS	LAST UPDATE
Mrs. Scarlett Wuckert V	182 Nolan Club	processing delivery	2023-10-05 11:16:06

Dashboard

Booking

Manifest

Delivery

Reports

Shipper

Consignee

Users

Merchandise

Sabin Bhurtel
User

Single Booking

Home » Single Booking

CN No

40000003

Date

10/10/2023

Category

Domestic

Payment Mode

☒ Cash ☐ Credit

Shipper

Choose Shipper

Create new shipper

Shipper

Choose Consignee

Create new consignee

Number

Address

Enter an address here

Number

Address

Enter an address here

Content Type

Doc

Merchandise Type

Choose Merchandise

Mode

Surface

Quantity

0

Weight

0

Individual Price

Price Before Discount

Discount

Price After Discount

Bill

Dashboard

Booking

Manifest

Delivery

Reports

Shipper

Consignee

Users

Merchandise

Sabin Bhurtel
User

Master Booking

Home » Master Booking

Booking Code

Search by number

Consignee

Choose Consignee

Sender

Choose Sender

Date

mm/dd/yyyy

Shipper Location

Search by Location

Consignee Location

Search by Location

Shipper Number

Search by number

Consignee Number

Search by Number

Search

BOOKING CODE	BOOKED	CONSIGNEE	DESTINATION	PHONE	STATUS	ACTIONS
40000001	2023-08-10	Mrs. Scarlett Wuckert V	182 Nolan Club	(628) 913-8804	processing delivery	View
40000002	2023-10-05	Sid Schneider	25007 Hoeger Center	+1.843.529.5300	processing delivery	View

Dashboard

Booking

Manifest

Delivery

Reports

Shipper

Consignee

Users

Merchandise

Sabin Bhurtel
User

Create Delivery

Home » Create Delivery

Date

10/10/2023

Route

Choose Route

Mode

Choose Vehicle

CN No	Consignee	Address	Contact No	Booked On	Shipped On

Save Delivery



Dashboard

Booking

Manifest

Delivery

Reports

Shipper

Consignee

Users

Merchandise

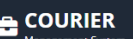
Everest Innovative Courier & Cargo Service

Kalimati, Kathmandu, Nepal | Tel: 01-5244145, 01-5244257
www.eiccs.com.np

Origin: EICCS KTM Route: Thankot Vehicle: Blke Date: 2023-10-05

S.NO	CN NO	CONSIGNEE	ADDRESS	CONTENT	QUANTATY	WEIGHT	SHIPPED DATE	STATUS
1	40000001	Prof. Jaylon Gleason	182 Nolan Club	porro	0	0	2023-10-05 11:16:06	Processin
2	40000002	Caesar Bradtke	25007 Hoeger Center	autem	1	1	2023-10-05 11:16:06	Processin

Delivery BY : Sabin



Dashboard

Booking

Manifest

Delivery

Reports

Shipper

Consignee

Users

Merchandise

Cash Report

Home » Cash Report

From To

BOOKING CODE	BOOKED	SENDER	DESTINATION	AMOUNT
40000002	2023-10-05	Dr. Marvin Altenwerth PhD	25007 Hoeger Center	23411.00
TOTAL				23411



Dashboard

Booking

Manifest

Delivery

Reports

Shipper

Consignee

Users

Merchandise

View

Home » View



Everest Innovative Courier & Cargo Service

Kalimati, Kathmandu, Nepal | Tel: 01-5244145, 01-5244257
www.eiccs.com.np

Sender: Mrs. Scarlett Wuckert V
Number: 678-467-1088
Address: 11430 Muller Port Apt. 505, Apt. 972

Receiver: Prof. Jaylon Gleason
Number: (628) 913-8804
Address: 182 Nolan Club, Apt. 945

Description:

Mailing Mode	Surface	Payment Mode	Cash
Declared Value	0	Quantity	0
Content	Doc	Weight	0
Booked On:	2023-08-10 AD	Amount:	Rs 0.00

Courier Consignment



40000001

TERMS AND CONDITIONS:

Unless a greater value is declared in writing in the space provided on this airway bill and freight on value charges are paid, the shipper declares and agrees that the declared value of the shipment is up to Rs.100, the shipper agrees that he has read and understands the terms and conditions.

Sabin Bhurtel

127.0.0.1:8000/booking/2/view

Everest Innovative Courier & Cargo Service
 Kalimati, Kathmandu, Nepal | Tel: 01-5244145, 01-5244257
 www.eiccs.com.np

Courier Consignment

 40000001

Sender: Mrs. Scarlett Wuckert V
 Number: **678-467-1088**
 Address: **11430 Muller Port Apt. 505, Apt. 972**

Receiver: Prof. Jaylon Gleason
 Number: **(628) 913-8804**
 Address: **182 Nolan Club, Apt. 945**

TERMS AND CONDITIONS:
 Unless a greater value is declared in writing in the space provided on this airway bill and freight on value charges are paid, the shipper declares and agrees that the declared value of the shipment is up to Rs.100, the shipper agrees that he has read and understands the terms and conditions.

Description:

Mailing Mode	Surface	Payment Mode	Cash
Declared Value	0	Quantity	0
Content	Doc	Weight	0
Booked On:	2023-08-10 AD	Amount:	Rs 0.00

Sabin Bhurtel
 For: EICCS

Received with thanks NRs 0.00. In Words zero

Print
2 pages

Destination
Save as PDF

Pages
All

Layout
Landscape

More settings

Save Cancel

COURIER
 Management System

Dashboard
Booking
Manifest
Delivery
Reports
Shipper
Consignee
Users
Merchandise

Create Manifest
 Home > Create Manifest

Date
10/10/2023

Location
Apt. 204

Mode
Choose Mode

CN No	Consignee	Address	Contact No	Booked On
Choose File Screen... (11).png 40000002	Caesar Bradtke	25007 Hoeger Center	+1.843.529.5300	2023-10-05
Choose File No file chosen				

Save Manifest